Sheriff case SHF-20261023-7b9227

This case ran `all` at the deep tier on 2026-10-23. It sweeps the three repositories on `origin/main` and adds five independent readers on the claims that carried weight among what merged since SHF-20261017-e4d7f6. It is an audit record: it changes no research state and repairs nothing. The verdicts are in `findings.yaml`, which passes `check-case`. The full text is in `REPORT.md`.

Result: of 16 claims, 2 are upheld, 10 are upheld with a narrower scope, 3 are overclaimed and 1 is broken. There is no blocker. The goal closure (GOAL-MONO-001) holds: its mathematics survived a blind re-derivation and an exhaustive falsification search. The broken claim is that `validate` guards main: it fails on every main run.

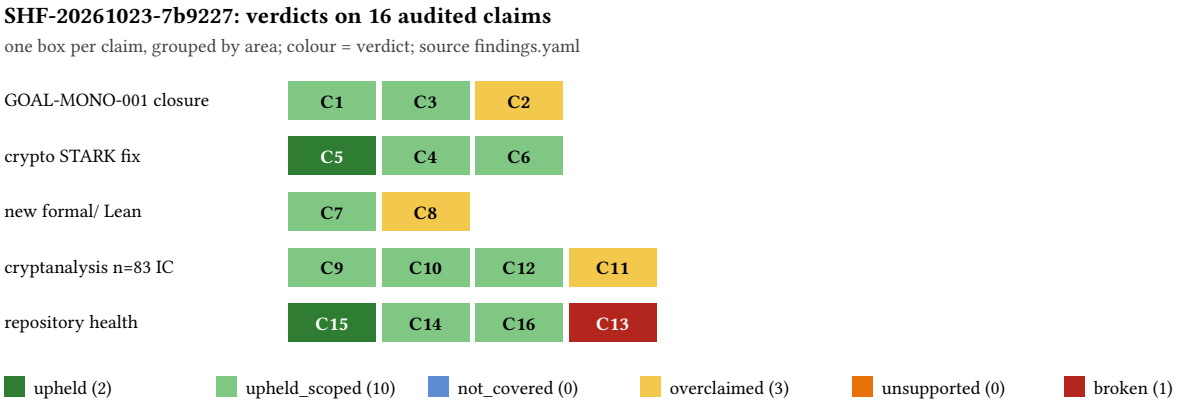


Figure 1: Verdicts on the 16 audited claims, by area (findings.yaml).

Scope

repository	commit	how
crypto-autoresearcher	`e9b91d62a9` origin/main	in place
crypto	`6ed792e86` origin/main	sparse partial clone; `ecbench` rebuilt
cryptanalysis	`6dfc25d` origin/main	sparse partial clone

- The canary passed (15/15, no false positives). The instrument includes the I1 fixes, so sparse coverage now reports as partial: 413 leads are `not_covered`.
- Independence: this session's own #2330 and the earlier cases are excluded. The STARK fix `999d4eefc` and #2326 were written by other sessions and are in scope.
- About 4.4 GB of disk was free, so only the Mathlib-free Lean modules could be built.

Prior (`prior.md`, written before the sweep):
- the STARK fix would hold, with risk in the AIR identifier — confirmed, minor;
- the ledger would have grown — it shrank, 157 ⟶ 141;
- CI would still be off — partly on, and red;
- new strong wording — confirmed;
- the cryptanalysis K-N residues would still be open — mostly fixed.

Verdicts

id	claim	verdict and basis
C1	KN-FIND-bdcd60 monodromy and Frobenius rule	upheld_scoped: blind re-derivation; orders 2/24/192/1920/23040; 0 violations in 2.1 M fibres; m≥4 for primitivity
C2	"established; three derivations agree"	**overclaimed**: one derivation, one blind (answer-shaped card), one guided; one model family
C3	GOAL-MONO-001 completed	upheld_scoped: criterion (c) instrument no-go; question open
C4	STARK statement binding; forgeries rejected	upheld_scoped: every old mutant rejected, no panics, 41/41 tests; new-design attacks not run
C5	stark_v2 soundness text	upheld
C6	ecbench `records_bound` (E1-R)	upheld_scoped: correct; nothing consumes it
C7	21 new formal/ Lean modules	upheld_scoped: no holes; constants re-derive; 6 Mathlib-free modules built here, 30 theorems standard axioms, mutants fail; 15 never built by CI
C8	"the certificate proves C5"	**overclaimed**: two C5 steps are Sage checks, not Lean

C9	n=83 closure 6.44 / 7.66 / 7.41	upheld_scoped: re-derived blind; 174/174 answers; gain is fewer unknowns, not cheaper relations
C10	n=83 engineering chain	upheld_scoped: wall time on four CPUs; sub-2% effects not separable from layout
C11	n=83 row: gap 3.9242, band not computable	**overclaimed**: [3.16, 5.22], host-dependent; band computable, unranked
C12	#944 fixes K-N1..4, R1-R	upheld_scoped: K-N1 partly (no group-op counts)
C13	`validate` guards main	**broken**: fails on every main run; 141 ledger errors
C14	runbook sessions audited	upheld_scoped: 2 unaudited; both audited clean here
C15	crypto #2066 L-N1, L-N7	upheld: the cited CI jobs re-read, all green
C16	fb-archive index	upheld_scoped: 2,681 archives, 0 problems

Major findings (no blocker)

- **MN-3:** the MONO confidence wording overstates the independence of the review round. The mathematics stands.
- **LN-1:** the FROB C5 report says Lean "proves C5". Two of its steps are checked in Sage, where they hold.
- **N83-1:** each relation costs one generic table solve, so the closure's gain comes from 600 $\longrightarrow$ 16 unknowns. As $K' \longrightarrow 0$ the method becomes the ledger's own BSGS on the quotient, which is not IC. The fix is to add a known-log arm.
- **CI-1:** `validate` is red on every main run, so it cannot flag new defects.
- **CI-2:** `lean-verification` is still disabled, so CI has built none of the 21 new Lean modules.

There are 18 minor findings; they are listed in `REPORT.md` and `findings.yaml`.

Sweep

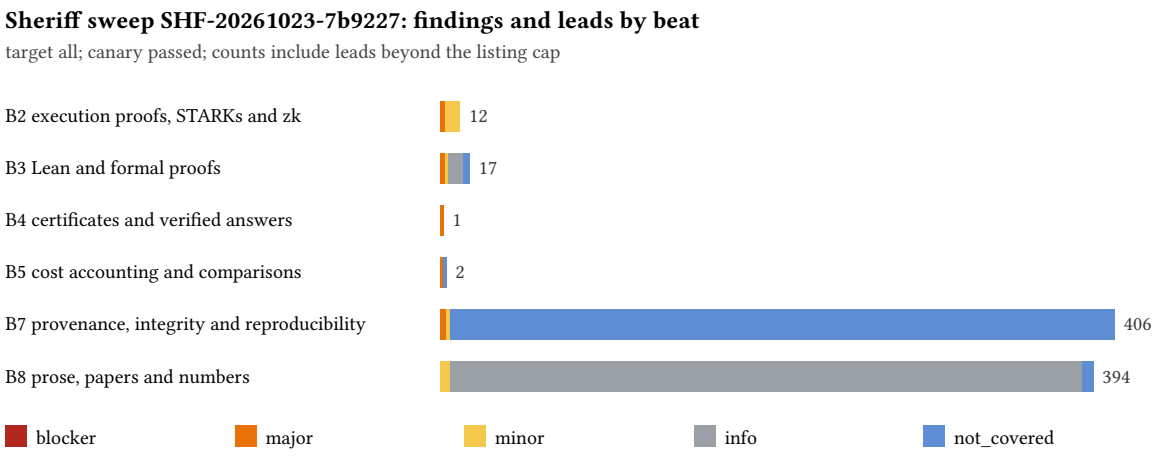


Figure 2: Deterministic sweep leads by beat (sweep.json).

Not covered

- **The 15 Mathlib-dependent Lean modules:** need about 6 GB of disk, or `lean-verification` re-enabled.
- **The STARK attacks on the new design:** AIR-identifier collision and oversized statements.
- **The 413 sparse-coverage leads, `refs.py`, `curve-names`, `ecbench bound check`, and the 497 ecbench sessions outside the sparse set:** need full checkouts.
- **Literature dominance for KN-FIND-bdcd60.**

What this case does not establish

- That KN-FIND-bdcd60 is new or bears on relation rates.
- That n=83 IC beats rho: it is 3.92× above matched rho, and the two are unranked across the GAE band.
- That the 15 Mathlib-dependent modules build.
- Anything about this session's own #2330.
- Any execution proof.

Readers (paths under `/tmp/...` in the reader files refer to the staging copy mirrored under `scripts/readers/`):

- validator-breakthrough: MONO, blind re-derivation;
- red-team-breakthrough (claude-fable-5-1): MONO, scope and decision logic;
- validator: STARK;
- validator: Lean;
- red-team: n=83.